4(a)	On the surface of the Earth, $\phi = -6.2\times10^{7}$ J kg^{-1} $\phi = -\dfrac{(6.67\times10^{-11})M}{6.4\times10^{6}} = -6.2\times10^{7}$ $M = 5.95\times10^{24}$ kg	**[1] Φ** **[1] ans**
(b)(i)	At distance R above the Earth's surface, $x = 2R$ $\phi = -3.2\times10^{7}$ J kg^{-1} By conservation of energy, U loss $= KE$ gain $m\left[0-(-3.2\times10^{7})\right] = \dfrac{1}{2}mv^{2}$ $v = 8000$ m s^{-1}	**[1] Φ** **[1] sub** **[1] ans**
(ii)	speed / velocity / acceleration would be greater. Deviates / bends from straight path	**[1]** **[1]**
5(a)	Electric field strength at a point is defined as the **electric force per unit positive**	**[1]**